

Facundo Gil

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Profile

Electrical Engineer with experience in data processing, automation, and data analysis. Interested in power system simulation, climate and energy forecasting, and energy analytics.

Experience

Ventus Ing.

Jul 2026 – Present

Data Engineer

Montevideo, Uruguay

- Design and development of Python pipelines to automate the extraction, transformation, and consolidation of financial and cost control data across multiple projects.
- Migration of legacy processes in R, Excel and Power Query to automated pipelines, standardizing data sources and improving performance, maintainability, and reusability.
- Integration of data from SAP Business One / SAP HANA and design of consolidated data models for Power BI, reducing manual processes and centralizing reporting across projects.

UdelaR – IMERL

Mar 2026 – Present

Grade 1 Teaching Assistant, Discrete Mathematics

Montevideo, Uruguay

- Taught practical classes for undergraduate students in discrete mathematics, strengthening communication, formal reasoning, and technical mentoring skills.

UTE – Administración Nacional de Usinas y Transmisiones Eléctricas

Sep 2024 – Sep 2025

Supply Planning Analyst

Montevideo, Uruguay

- Developed Python automation tools to process and validate outputs from large-scale power flow simulations, reducing manual data preparation and analysis work.
- Designed repeatable workflows to transform raw PSSE/E simulation outputs into structured datasets for grid expansion studies and economic planning.
- Analyzed high-volume grid expansion simulation results to support infrastructure planning and investment decisions.
- Generated technical and economic reports from processed simulation data for internal planning teams and decision-makers.
- Analyzed meteorological data for energy simulations.

ISBEL S.A

Jun 2023 – Sep 2024

Support Assistant

Montevideo, Uruguay

- Provided support for production infrastructure across VMware virtualized environments, network systems, VoIP platforms, and database services.
- Performed disk hardening, infrastructure diagnostics, system configuration, and preventive maintenance in client environments.
- Resolved high-priority incidents related to virtualization, connectivity, service availability, and backend infrastructure components.

Education

Databricks Engineer Associate

2026

Databricks

- Databricks Data Engineer Associate Certification.

Universidad de la República (UdelaR)

2018 – 2025

Electrical Engineering – Signal Processing and Machine Learning

- **Relevant coursework:** Graph Machine Learning, Numerical Linear Algebra and HPC, Power System Simulation, Machine Learning Fundamentals.

Cambridge Proficiency in English (Cambridge University)

2018

Cambridge University International English Examination

- C2-level English examination in speaking, listening, reading and writing.

Energy Systems Simulation

2025

UdelaR Graduate Course

- Course focused on power generation system simulation.

Related Projects

YouTube Trending Pipeline | *Amazon Web Services - Python*

2026

- Built an end-to-end, event-driven ETL pipeline deployed on AWS to analyze trends among the most popular YouTube videos in Uruguay.
- Implemented ingestion, processing, and storage layers using AWS Lambda, S3, CloudWatch, Redshift, and dbt.
- Used Terraform for infrastructure as code.

MPC Control and PINNs | *PyTorch, ROS2, Gazebo*

Jun 2024 – Jun 2025

- Engineering capstone project: implemented physics-informed neural networks and a model predictive control pipeline for a 6-degree-of-freedom robotic arm simulated in ROS2/Gazebo.
- Used ClusterUy for distributed computing resources during training.

GNN for Power Flow Prediction | *PyTorch, Graph ML*

2025

- Tested multiple graph neural network architectures for power flow prediction in transmission networks.

For more details about my projects, visit my website: facug96.github.io

Skills

Languages : Python, SQL, Bash, JavaScript, R, C/C++

Data Engineering : Airflow, dbt, PySpark, PostgreSQL, Databricks, Delta Lake, pandas, NumPy

Cloud and Infrastructure : AWS (Lambda, S3, EventBridge, Redshift, CloudWatch, IAM), Terraform, Docker, Linux

BI and Tools : Power BI, Git, GitHub, Jupyter, SQLAlchemy